



Bayan

Simplify Arabic where you read
on the phone, offline

Team Cogni

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Reading in Arabic adds load

11%

of Arab primary-school children
have developmental dyslexia

Al-Dakhil 2024 · 18 studies, N = 30,243



Reading in Arabic adds load

What helps

Splitting long clauses into short sentences: read faster, understood better, most by the weakest readers.

Javourey-Bonnet 2022

Bayan's scope

we reduce reading load

long clauses · hard words

tashkeel is a setting · no reader claim yet

How we got to corpus v1

SAMER · human-written

novels, short sentences

42–49% of pairs unchanged

length ratio 1.00: swaps words, never splits

Other sources · LLM-written

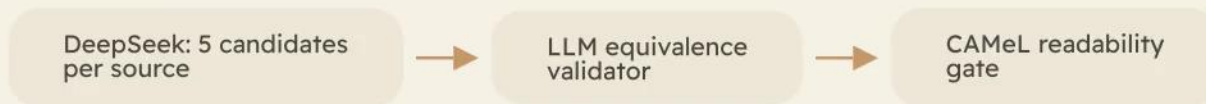
many targets are summaries

they drop facts, delete options

after a meaning filter: 6,888 pairs

How we got to corpus v1

corpus v0, our first synthetic data



What the audit found

57% of rows had tashkeel

63% of cut-off sources were simply completed

only 43% of pairs were 2+ levels easier

the validator accepted 34% of planted added sentences · we trained nothing on v0

How we got to corpus v1

V1 · FROM THE SOURCE

BAREC train + dev
test never enters



strip tashkeel and tatweel
every text



route each source
protected · short · hard → 23,221 hard

V1 · INTO THE CORPUS

Gemma 4 31B writes candidates
hard band only → 23,205 responses



code gates
numbers · quotes · options



readability + meaning gates
≥ 2 levels · meaning gate → 10,693 accepted

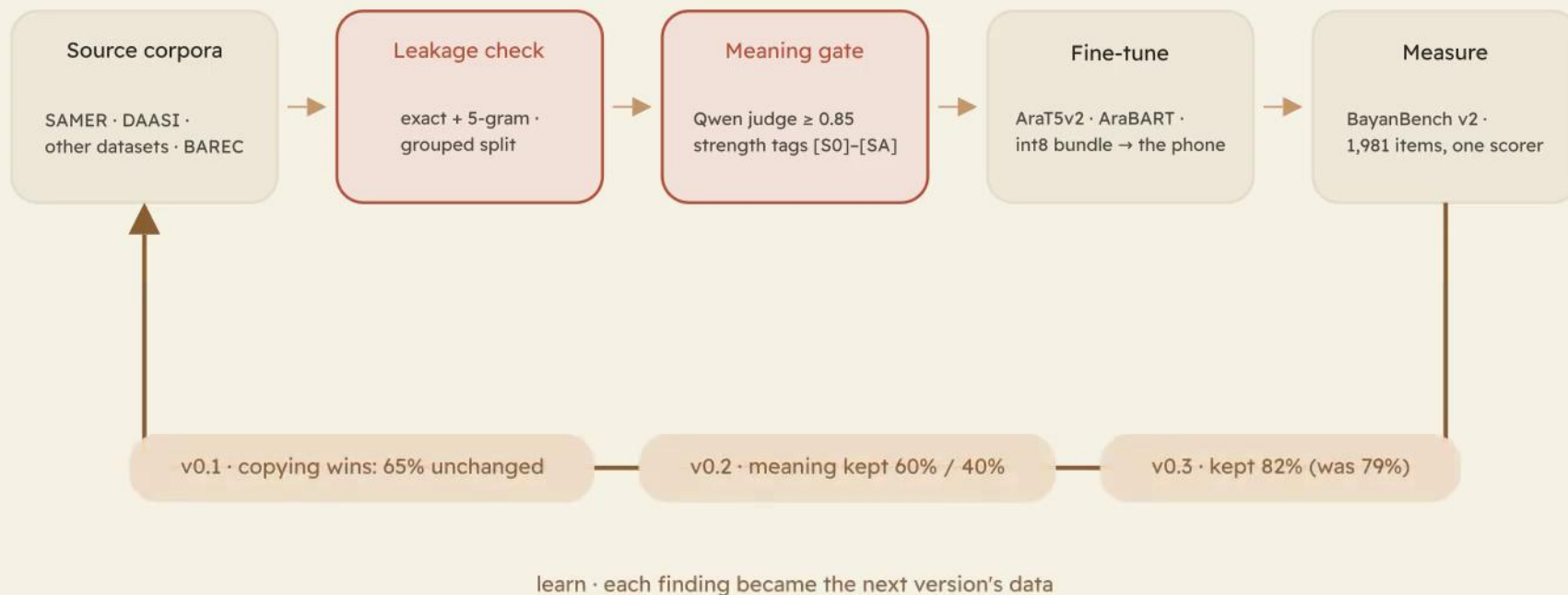
How we got to corpus v1



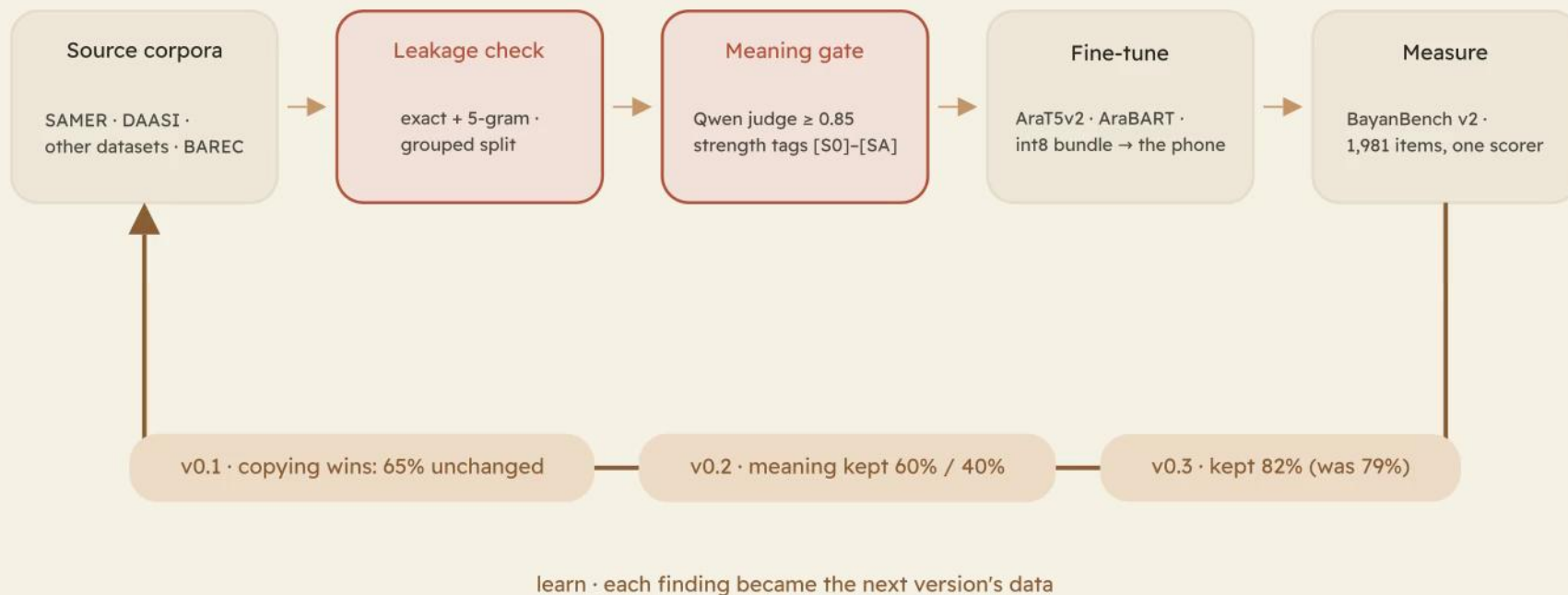
Build, measure, learn



Build, measure, learn



Build, measure, learn



Each version answered the last failure

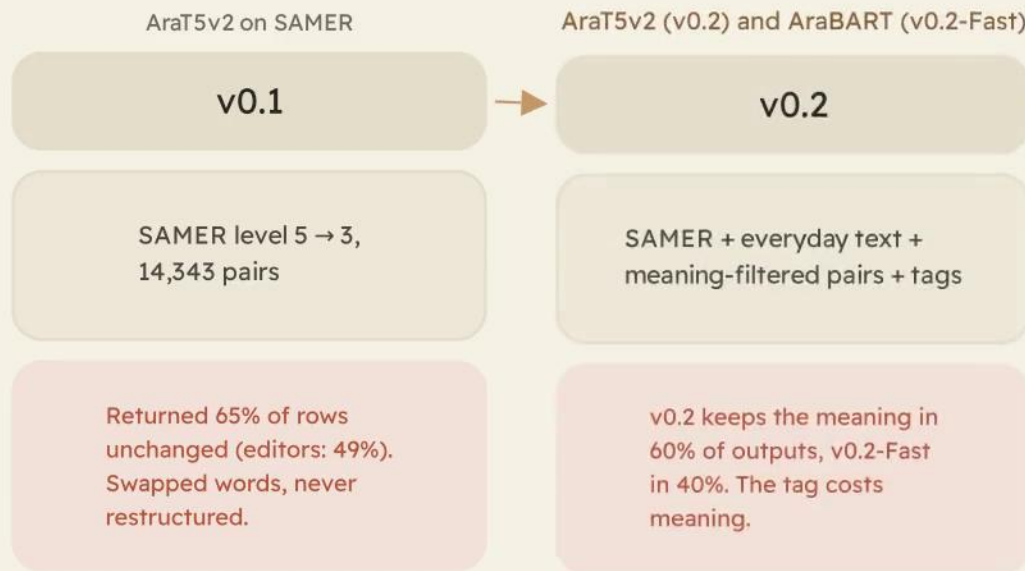
AraT5v2 on SAMER

v0.1

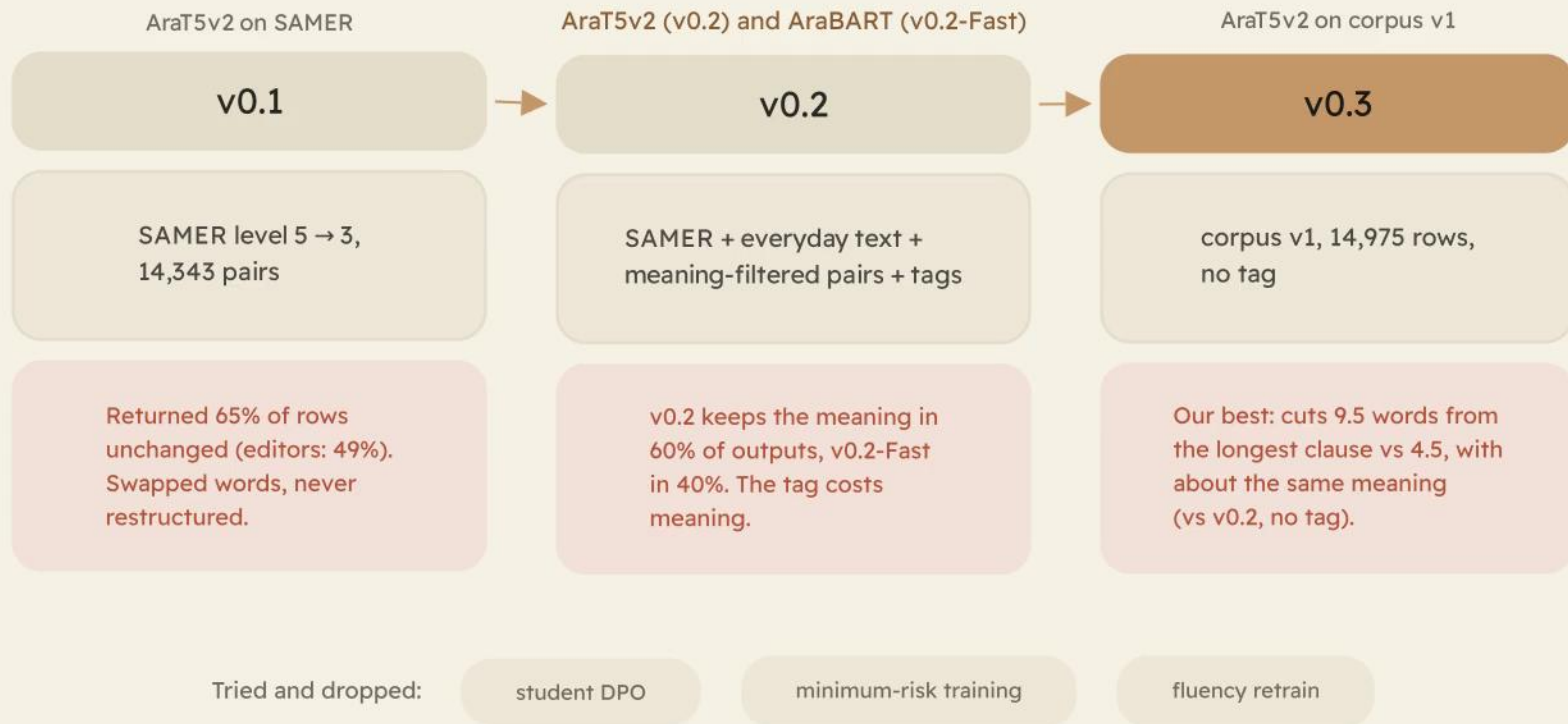
SAMER level 5 → 3,
14,343 pairs

Returned 65% of rows
unchanged (editors: 49%).
Swapped words, never
restructured.

Each version answered the last failure



Each version answered the last failure



We checked our own yardstick

Copying is the baseline to beat

	meaning kept	words cut from the longest clause
Copy the input	100%	0
v0.1	76%	0
v0.2 (AraT5v2)	68%	4.5
v0.2-Fast (AraBART)	51%	4.9
v0.3	69%	9.5

1,981
items, split by document

731 dev · 1,250 test

BayanBench v2

test core items, as trained (float, greedy) · the shipped app model with four beams is on the results slide

We checked our own yardstick



We checked our own yardstick

Meaning

Does it still say the same thing?
Same meaning, every number intact.

Simpler

Are long clauses shorter?
Words cut from the longest clause.

Restraint

Does it leave alone what should
stay? Easy text, scripture.

Copying rate

How often is the text returned
unchanged? A copy counts as
perfect meaning, so we report it.

We checked our own yardstick

0.85

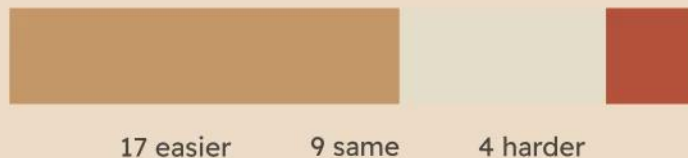
scorer vs human raters (AUC)
6 raters · 290 tasks

catches 42 of 51 changed outputs

fails 53 of 194 that people kept: stricter

raters barely agree on ease

A reader with dyslexia
rated 30 outputs



a sample of one, not a study

v0.3 is our best model



BayanBench v2, test core items

the more it simplifies,
the less meaning it keeps, obviously

v0.3 keeps the most meaning (82%)
and still cuts 5.2 words off
the longest clause

v0.3 is the app's large model,
decoded with four beams

v0.3 is our best model

In the app: v0.3 (four beams) minus v0.2, same text step, test core, 95% interval



+18 points more easy text left unchanged · every number kept (100%)

v0.3 is our best model

Blind rating by the team: 18 sentences

outputs the system changed, rated easier to read

v0.3



86%

v0.2



60%

v0.2-Fast



45%

meaning kept in 96% of ratings (v0.2: 96%, v0.2-Fast: 92%) · none rated harder

8 of 8 team votes chose v0.3 over v0.2

the reader with dyslexia chose v0.3 in 4 of 4 comparisons it was in · 18 sentences: a direction, not a size

What the app does

Select Arabic text in any app.

Tap tabseet. Bayan simplifies it
and reads it aloud.





ويكيبيديا



تمدد عمراني



التمدد العمراني أو الزحف العمراني هو مفهوم عام متعدد الوجوه يُشير إلى توسع **مدينة** ما **وضواحيها** على حساب الأراضي والمناطق المحيطة بها. تؤدي هذه الظاهرة إلى تطوير المناطق الريفية المجاورة للمدن الكبيرة تدريجياً وزيادة **كثافتها السكانية** شيئاً فشيئاً، كما تُساعد على رفع مستوى الخدمات فيها وتتسبب بانتشار استخدام **السيارات** ووسائل النقل الحديثة بها على مستوى أوسع.^[1]



مثال على الزحف العمراني أو التمدد العمراني في مدينة سان خوسيه بكاليفورنيا.



What the app does

Select Arabic text in any app.

Tap tabseet. Bayan simplifies it
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Simplify where you read.



Simplify where you read.

no reader has used Bayan; one reader rated 30 outputs

our scorer is stricter than people

the comparison with people covers 18 sentences

Next: a reading study with more readers.

Thank you · Team Cogni · Samsung Innovation Campus